

Chapter 9: Medians and Order Statistics (CORMEN)

(1)

1) $\rightarrow$ The i th order statistic of a set of n elements is the i th smallest element. For a set of n elements, minimum is the 1st order statistic and maximum is the n th order statistic.

The selection problem is to identify the i th order statistic of a set of n numbers.

$\rightarrow$

MINIMUM(A)

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1 min  $\leftarrow A[1]$ 
2 for  $i \leftarrow 2$  to length[A]
3   do if  $\text{min} > A[i]$ 
4     then  $\text{min} \leftarrow A[i]$ 
5 return min
```

Minimum or maximum can be determined with $n-1$ comparisons.

At most $3\lfloor n/2 \rfloor$ comparisons are sufficient to find both the minimum and the maximum. The strategy is to maintain the minimum and the maximum elements seen thus far. We process the elements in pairs. Pairs of elements from the IP are first compared with each other. Next the smaller of the two is compared with the smallest and the larger of the two with the largest, at a cost of 3 comparisons for every 2 elements.

If n is odd we set both the ^{initial} minimum and maximum to the value of the first element. If n is even, we compare the 1st 2 elements to determine the initial minimum and maximum.